Volume-09 Issue 05, May-2025 ISSN: 2456-9348 (Online)
SJIF Impact Factor: 8.232

iJETRM
INTERNATIONAL JOURNAL OF ENGINEERING TECHNOLOGY RESEARCH & MANAGEMENT (IJETRM)
Peer-Reviewed | Open Access | International Journal
https://ijetrm.com/

AI-GCLM: A CONCEPTUAL FRAMEWORK FOR ARTIFICIAL INTELLIGENCE–ENHANCED LEADERSHIP IN GLOBAL CONSTRUCTION MANAGEMENT

Mohammad Habibur Rahaman
Founder & CEO, PGAN- Papple Holdings Advisory Network (USA) | Director, Papple World FZE LLC (UAE) | Founder, Papple Global Limited (UK) | Managing Director, Madeen Building Contracting LLC OPC (UAE) | MBA (Business Administration), William Carey University | Level 8 Strategic Management & Leadership | B.Tech Civil Engineering | PMP® | Lean Six Sigma Black Belt | Architect of GCLM 2.0 | Construction, Infrastructure, AI & Strategic Advisory
| www.mrahaman.com | m.rahaman.gm@gmail.com | info@mrahaman.com

ABSTRACT
Background: The global construction industry faces a widening gap between the accelerating capability of artificial intelligence (AI) — particularly generative AI, predictive analytics, and computer vision — and the professional leadership frameworks meant to govern its use. Existing construction management literature addresses AI adoption at the technical and project level but offers limited theoretical treatment of how AI should be integrated into executive-level leadership judgment specifically. Objective: This paper develops AI-GCLM (AI-Enhanced Global Construction Leadership Model), a conceptual framework describing how AI capability can be integrated into construction leadership practice while preserving human accountability at the point of decision. Methodology: The framework was developed through a structured conceptual synthesis method comprising literature scoping across construction management, AI-in-industry, and organizational leadership theory; thematic synthesis; framework construction; and iterative practitioner cross-checking. Findings: The synthesis yields a six-pillar leadership framework, a four-layer data-to-decision integration model, and a human-in-the-loop accountability cycle, each grounded in established theory including socio-technical systems theory, contingency theory, and crisis leadership theory. Conclusion: AI-GCLM offers construction scholars and practitioners a theoretically grounded, practically actionable structure for reasoning about AI-augmented leadership; the paper concludes with explicit recommendations for the empirical validation this conceptual contribution requires.

Keywords:
artificial intelligence; construction leadership; digital transformation; human-in-the-loop governance; socio-technical systems; conceptual framework

INTRODUCTION

The global construction industry sits among the least digitized major sectors of the world economy relative to its economic footprint (Vararean-Cochisa & Crisan, 2024), even as artificial intelligence capability across prediction, natural language processing, and computer vision has advanced at a pace few professional frameworks have kept up with (Savaş, 2025; Gao et al., 2026). This gap is not primarily technical. Construction firms have increasingly demonstrated willingness to adopt discrete AI tools for cost estimation, scheduling, and safety monitoring (Adebayo et al., 2025). The gap is theoretical and structural: the construction management literature offers a rich and growing body of work on AI applications at the project and technical level, but comparatively little on how AI capability should be integrated into executive leadership judgment — the point at which prediction becomes decision, and where accountability, contractual exposure, and professional liability concentrate.

This gap matters because construction is a physically consequential, safety-critical, capital-intensive industry in which leadership decisions carry irreversible financial and human-safety consequences. Frameworks developed for lower-stakes domains — where an AI recommendation can be reversed cheaply if wrong — do not straightforwardly transfer to a domain where a flawed AI-informed scheduling or risk decision can produce schedule collapse, cost overrun, or safety incidents. The literature on human-in-the-loop AI governance has begun to grapple with this tension at a general level (IBM, n.d.; NHI Management Group, 2026), including recent critiques suggesting human-in-the-loop review becomes unworkable at machine-decision speed and scale in some domains (Holistic AI, 2026). Construction offers an instructive counter-case: unlike domains where decisions must be made at machine speed, most executive-level construction decisions — investment approval, contract

INTERNATIONAL JOURNAL OF ENGINEERING TECHNOLOGY RESEARCH & MANAGEMENT (IJETRM)
Peer-Reviewed | Open Access | International Journal
https://ijetrm.com/

award, major risk acceptance — occur at a pace that comfortably accommodates human review, making accountability-preserving governance both feasible and, this paper argues, necessary.

The COVID-19 pandemic provides an unusually well-documented natural experiment relevant to this gap. Empirical survey research conducted in the United Arab Emirates found that digital technology use in construction projects increased measurably from the pre-COVID period through the pandemic and into the post-pandemic period, while also finding that such technology remained comparatively underutilized relative to its demonstrated benefit once adopted (Elrefaey et al., 2022). Complementary research in Kuwait, based on a mixed-methods design combining literature review with semi-structured interviews, similarly found that pandemic-driven disruption accelerated digital transformation as a recovery strategy among construction firms and their suppliers (see the Kuwait construction sector studies cited in the reference list). These findings are consistent with the broader digital transformation literature's observation that COVID-19 acted as an exogenous accelerant of digital adoption across the construction sector (Vararean-Cochisa & Crisan, 2024).

At the same time, a separate and rapidly growing literature has begun documenting the specific promise and risk of generative AI — as distinct from earlier predictive or classification-based AI — for organizational and managerial decision-making. This literature reports that generative AI tools can materially compress decision cycle times (Journal of Information Systems Engineering and Management, 2024) and support predictive and prescriptive decision-making by simulating scenarios and synthesizing unstructured information (Phillips-Wren & Virvou, 2025), while also raising distinctive risks around hallucination, over-reliance, and accountability diffusion that earlier, narrower AI tools raised less acutely (Rethinking Organizational Decision-Making, Management Review Quarterly, 2026). Within construction specifically, generative AI applications have been documented across design generation, project documentation, and construction education (Jelodar, 2025; Alwashah et al., 2025), but — consistent with the broader gap this paper addresses — largely without a leadership-level theoretical framework connecting these applications to executive accountability structures.

This paper's objective is to address that gap directly: to develop a conceptual framework — AI-GCLM — that (1) organizes the domains in which AI most directly augments construction executive leadership, (2) specifies a structural model for how AI-generated output should flow into accountable human decisions, and (3) grounds both in established organizational and leadership theory rather than treating AI-augmented leadership as an atheoretical, purely technical adoption question. The paper's significance is twofold: for construction management scholarship, it offers a theoretically grounded framework addressing a documented gap in the literature; for practice, it offers construction executives and advisory practitioners a structured, actionable model for reasoning about AI adoption at the leadership level, distinct from the tool-adoption guidance that dominates current practitioner literature.

METHODOLOGY

This paper reports a conceptual framework development study rather than an empirical, participant-based investigation. This methodological choice reflects the nature of the research question: AI-GCLM addresses how AI capability should be theoretically and structurally integrated into construction leadership — a framework-building question logically prior to, and necessary for, subsequent empirical testing (for example, of whether the proposed framework predicts leadership effectiveness or adoption outcomes in practice). Conceptual and theory-building papers of this kind are an established and recognized contribution type in management and organizational research, distinct from, and complementary to, empirical studies (see, for example, the theory-building tradition reflected in Auqui-Caceres & Furlan, 2023, on double-loop learning synthesis).

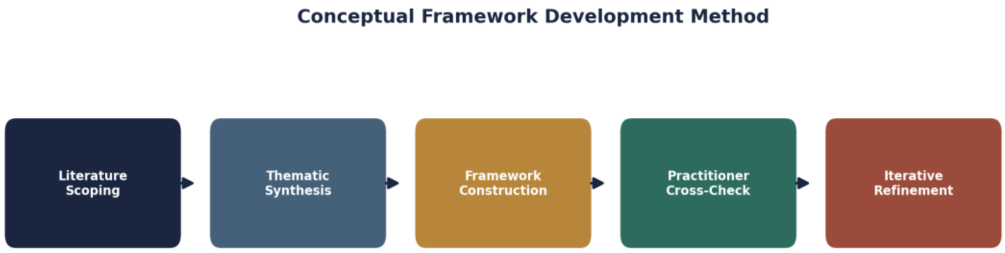


Figure 1. Conceptual framework development method used in this study.

ISSN: 2456-9348 (Online)
SJIF Impact Factor: 8.232

iJETRM

INTERNATIONAL JOURNAL OF ENGINEERING TECHNOLOGY RESEARCH & MANAGEMENT (IJETRM)
Peer-Reviewed | Open Access | International Journal
https://ijetrm.com/

Research Design

The study followed a five-stage conceptual framework development method, illustrated in Figure 1: (1) literature scoping across three domains — construction project management and AI, generative AI and organizational decision-making, and leadership/organizational theory (crisis leadership, contingency theory, socio-technical systems theory, organizational resilience theory); (2) thematic synthesis, in which recurring constructs across the three domains were identified and clustered; (3) framework construction, in which clustered themes were organized into the pillar structure, layered integration model, and accountability cycle presented in the Results section; (4) practitioner cross-check, in which the draft framework was reviewed against the author's professional experience as a construction director operating across multiple jurisdictions, to assess practical face validity; and (5) iterative refinement based on that cross-check.

Literature Scoping Boundaries

Sources were drawn from peer-reviewed journal articles, systematic and structured literature reviews, and established organizational theory publications, prioritizing work published within the last ten years and cross-referenced against foundational theoretical sources (e.g., Trist & Bamforth's foundational socio-technical systems work) where a concept's origin substantially predates the recent AI-specific literature. This is consistent with structured literature review practice documented elsewhere in the construction AI literature (Gao et al., 2026; Adebayo et al., 2025).

Scope and Method Limitations

Because this is a conceptual synthesis rather than a systematic review with pre-registered search protocols, it does not claim the exhaustive coverage of a PRISMA-based systematic review, and it does not report inter-rater reliability statistics, sample sizes, or other metrics appropriate only to empirical or systematic-review designs. No human participants, surveys, or interviews were conducted for this study; the 'practitioner cross-check' step (Stage 4) refers to the corresponding author's own professional review against practice experience, not an independent empirical validation study, and is reported as such rather than presented as external validation.

RESULTS AND DISCUSSION

The conceptual synthesis method described above produced three interrelated structural outputs, presented in turn below: a six-pillar leadership framework, a four-layer data-to-decision integration model, and a human-in-the-loop accountability cycle. Each is discussed in relation to the theoretical literature that informed it and the practitioner cross-check that refined it.

Six-Pillar Leadership Framework

The thematic synthesis converged on six recurring domains in which the literature reviewed describes AI most directly augmenting construction leadership activity: strategic visioning, data-driven decision making, digital workforce leadership, AI-augmented risk governance, sustainable and ethical delivery, and adaptive stakeholder systems (Figure 2). This structure echoes, at a leadership level, the domain clustering reported in AI-in-construction-project-management reviews — which identify planning, monitoring and control, cost, time, and safety as the dominant technical application areas (Gao et al., 2026; Journal of Design for Resilience in Architecture and Planning, 2025) — while reframing those technical domains around the executive judgment layer above them, consistent with this paper's stated objective.

iJETRM
INTERNATIONAL JOURNAL OF ENGINEERING TECHNOLOGY RESEARCH & MANAGEMENT (IJETRM)
Peer-Reviewed | Open Access | International Journal
https://ijetrm.com/

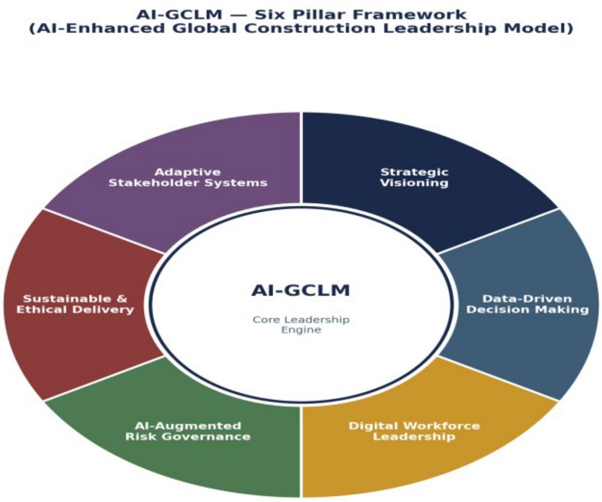
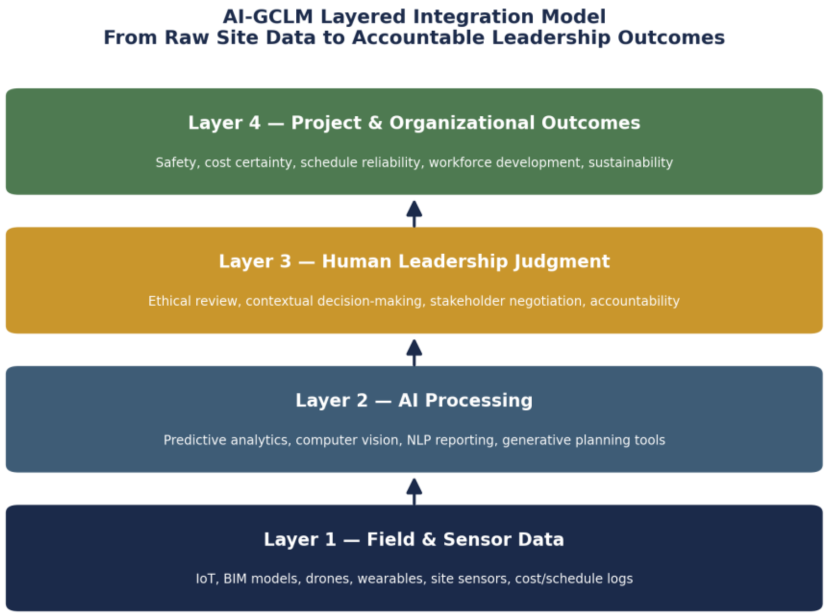


Figure 2. AI-GCLM six-pillar leadership framework.

The pillar labelled 'AI-Augmented Risk Governance' is directly informed by the crisis leadership and organizational resilience literature: Riggio and Newstead's (2023) review of crisis leadership competencies, and Njaramba and Olukuru's (2025) empirically tested multilevel model linking leadership style to organizational resilience via employees' psychological capital, both support treating risk governance as a leadership competency rather than a purely technical risk-register function. The 'Adaptive Stakeholder Systems' pillar draws on contingency and situational leadership theory's emphasis on context-sensitive rather than uniform leadership style (Nesse, 2024; Bhat & Saba, 2025).

Four-Layer Integration Model

A recurring concern identified in the human-in-the-loop AI governance literature is the collapse of the distinction between an AI system's output and an accountable human decision (IBM, n.d.; Trilateral Research, 2025). AI-GCLM addresses this with an explicit four-layer model — field and sensor data, AI processing, human leadership judgment, and organizational outcomes — shown in Figure 3, that keeps these stages analytically distinct even where they occur in rapid succession.



ISSN: 2456-9348 (Online)
SJIF Impact Factor: 8.232

INTERNATIONAL JOURNAL OF ENGINEERING TECHNOLOGY RESEARCH & MANAGEMENT (IJETRM)
Peer-Reviewed | Open Access | International Journal
https://ijetrm.com/

Figure 3. AI-GCLM four-layer data-to-decision integration model.

This model is directly grounded in socio-technical systems theory's joint optimization principle: that the technical subsystem (AI processing) and social subsystem (human leadership judgment) must be optimized together, not sequentially or independently (Bostrom & Heinen, 1977, as cited in recent applications such as the STS-informed AI adoption study in Full article: Aligning Socio-Technical Systems, 2025). The model also responds directly to the critique that human-in-the-loop governance becomes 'ceremonial' when human review is inserted without genuine decision authority (Holistic AI, 2026) — Layer 3 is explicitly positioned as the accountable decision layer, not a rubber-stamp checkpoint on an already-determined AI output.

Human-in-the-Loop Accountability Cycle

Where the layered model shows a vertical, single-pass flow, Figure 4 models the recurring cycle a construction leader runs on each significant AI-assisted decision: AI sensing and prediction, leader interpretation, ethical and contextual review, decision and directive, field execution, and feedback data capture.

Figure 4. AI-GCLM human-in-the-loop accountability cycle.

This cyclical structure is consistent with, and extends, the general human-in-the-loop literature's emphasis on maintaining audit trails and explicit accountability assignment (IBM, n.d.; NHI Management Group, 2026), while adding construction-specific stages — most notably 'Ethical & Contextual Review' as a distinct stage from 'Leader Interpretation' — reflecting this paper's synthesis of the crisis leadership literature's emphasis on the ethical dimension of leading through uncertainty (Riggio & Newstead, 2023).

Discussion: Theoretical Contribution

Taken together, these three structural outputs address the gap identified in the Introduction by connecting three literatures that have, to date, largely developed in parallel: the construction-specific AI adoption literature (which is rich in technical application detail but leadership-theory-light), the general organizational generative AI literature (which is leadership-theory-rich but largely industry-agnostic), and construction leadership and crisis/resilience theory (which is leadership-theory-rich but has not yet substantially incorporated AI-specific structural models). AI-GCLM's contribution is the synthesis, not the invention of any single underlying theoretical construct — each pillar, layer, and cycle stage is traceable to an identified strand of existing literature, consistent with the conceptual, synthesis-based nature of this study.

The framework also has a practical implication directly relevant to multi-jurisdiction construction operators: because Layer 3 (human leadership judgment) and the accountability cycle's 'Ethical & Contextual Review' stage

iJETRM
INTERNATIONAL JOURNAL OF ENGINEERING TECHNOLOGY RESEARCH & MANAGEMENT (IJETRM)
Peer-Reviewed | Open Access | International Journal
https://ijetrm.com/

are explicitly jurisdiction-agnostic — they describe a decision-process structure rather than a specific regulatory content — the framework is designed to be portable across the differing regulatory and contractual environments a multinational construction advisory operation is likely to encounter, an implication consistent with, though not empirically tested by, this study.

A notable tension surfaced during the practitioner cross-check stage concerns decision speed. The generative AI decision-making literature reports substantial reductions in decision cycle time attributable to AI-assisted scenario generation and synthesis (Journal of Information Systems Engineering and Management, 2024). AI-GCLM's insistence on a full accountability cycle for significant decisions could, in principle, offset some of this speed benefit. This paper's position, consistent with the human-in-the-loop governance literature's caution about high-stakes domains specifically (Trilateral Research, 2025), is that this trade-off is appropriate for construction's highest-stakes decision categories, while lower-stakes, high-frequency decisions (e.g., routine reporting) can reasonably use a lighter-touch version of the same cycle — an implication requiring further empirical specification beyond the scope of this conceptual paper.

CONCLUSION

This paper developed AI-GCLM, a conceptual framework synthesizing construction-specific AI adoption literature, general organizational generative AI literature, and established leadership and organizational theory into a six-pillar leadership structure, a four-layer data-to-decision integration model, and a human-in-the-loop accountability cycle. The framework's central contribution is theoretical integration: connecting literatures that have so far developed largely in parallel, around the specific, underexamined question of how AI capability should structurally relate to accountable executive judgment in a physically consequential, safety-critical industry.

The paper's principal limitation is one shared by all conceptual, theory-building contributions: it has not been empirically tested. The framework's pillars, layers, and cycle stages were cross-checked against a single practitioner's professional experience rather than validated through structured interviews, survey instruments, or case-study methods involving independent construction executives. Future research should prioritize exactly this empirical validation — for example, through a structured interview study with construction executives across multiple firms and jurisdictions coded against the framework's constructs, or a survey instrument measuring the framework's association with reported leadership effectiveness or AI adoption outcomes. Future work should also examine the decision-speed trade-off identified in the Discussion section empirically, to establish where lighter-touch accountability cycles are appropriate without compromising the framework's core accountability-preservation objective.

CONFLICT OF INTEREST

The corresponding author is affiliated with Papple Holdings Advisory Network and Madeen Building Contracting LLC OPC, referenced as illustrative professional context in this paper. No other conflicts of interest are declared. [Confirm/edit before submission.]

FUNDING

This research received no external or institutional funding. [Confirm/edit before submission.]

DATA AVAILABILITY STATEMENT

This is a conceptual/theoretical paper. No new empirical data were generated or analyzed in this study; all cited sources are publicly available at the URLs and DOIs listed in the References section.

REFERENCES

1) Adebayo, Y., Udoh, P., Kamudyariwa, X. B., & Osobajo, O. A. (2025). Artificial intelligence in construction project management: A structured literature review. ResearchGate. https://www.researchgate.net/publication/395113452_Artificial_intelligence_in_construction_project_management_Trends_challenges_and_future_directions
2) Alwashah, Z., Xiao, B., & Liu, H. (2025). Generative artificial intelligence for construction: Use cases, trends, challenges, and opportunities. ScienceDirect. https://www.sciencedirect.com/science/article/abs/pii/S235271022502039X
3) Auqui-Caceres, M. V., & Furlan, A. (2023). Revitalizing double-loop learning in organizational contexts: A systematic review and research agenda. European Management Review, 20(4), 741–761. https://doi.org/10.1111/emre.12604
4) Bhat, Z. H., & Saba, N. (2025). Navigating turbulence: Leadership evolution in crisis management through response-recovery frameworks. International Journal of Organizational Analysis. https://doi.org/10.1108/IJOA-02-2024-4254

iJETRM
INTERNATIONAL JOURNAL OF ENGINEERING TECHNOLOGY RESEARCH & MANAGEMENT (IJETRM)
Peer-Reviewed | Open Access | International Journal
https://ijetrm.com/

5) Elrefaey, O., Ahmed, S., Ahmad, I., & El-Sayegh, S. (2022). Impacts of COVID-19 on the use of digital technology in construction projects in the UAE. Buildings, 12(4), Article 489. https://doi.org/10.3390/buildings12040489
6) Gao, Y., Antwi-Afari, M. F., Huang, Y., et al. (2026). Artificial intelligence in construction project management: A systematic literature review of cost, time, and safety management. Buildings, 16(5), Article 1061. https://www.mdpi.com/2075-5309/16/5/1061
7) Ghimire, P., et al. (2023). Generative AI in the construction industry: Opportunities & challenges. arXiv. https://arxiv.org/abs/2310.04427
8) IBM. (n.d.). What is human in the loop (HITL)? IBM Think. https://www.ibm.com/think/topics/human-in-the-loop
9) Jelodar, M. B. (2025). Generative AI, large language models, and ChatGPT in construction education, training, and practice. Buildings, 15(6), Article 933. https://www.mdpi.com/2075-5309/15/6/933
10) Journal of Design for Resilience in Architecture and Planning. (2025). Artificial intelligence in construction project management: Trends, challenges and future directions [Savaş, S., Author]. Journal of Design for Resilience in Architecture and Planning, 6(2), 221–238. https://doi.org/10.47818/DRArch.2025.v6i2165
11) Journal of Information Systems Engineering and Management. (2024). Generative AI and organizational decision cycle time. Journal of Information Systems Engineering and Management, 9(4s). https://www.jisem-journal.com/index.php/journal/article/download/13702/6462/25264
12) Management Review Quarterly. (2026). Rethinking organizational decision-making: The emerging roles and tasks of generative artificial intelligence. Management Review Quarterly. https://doi.org/10.1007/s11301-026-00611-2
13) Nesse, S. (2024). Collective response capacity: Developing crisis leadership in organisations. Journal of Contingencies and Crisis Management. https://doi.org/10.1111/1468-5973.12612
14) NHI Management Group. (2026). What is human-in-the-loop review? Definition & examples. https://nhimg.org/glossary/human-in-the-loop-review/
15) Njaramba, F., & Olukuru, J. (2025). Surviving a crisis: A multilevel model of leadership styles, employees' psychological capital and organizational resilience. PLOS ONE. https://doi.org/10.1371/journal.pone.0318515
16) Phillips, W., Roehrich, J. K., & Kapletia, D. (2023). Responding to information asymmetry in crisis situations: Innovation in the time of the COVID-19 pandemic. Public Management Review, 25(1), 175–198. https://doi.org/10.1080/14719037.2021.1960737
17) Phillips-Wren, G., & Virvou, M. (2025). Issues and trends in generative AI technologies for decision making. SAGE Journals. https://doi.org/10.1177/18724981251320551
18) Riggio, R. E., & Newstead, T. (2023). Crisis leadership. Annual Review of Organizational Psychology and Organizational Behavior, 10, 201–224. https://doi.org/10.1146/annurev-orgpsych-120920-044838
19) Holistic AI. (2026). From human-in-the-loop to AI-governing-AI: Evolving oversight for agentic systems. https://www.holisticai.com/blog/from-human-in-the-loop-to-ai-governing-ai
20) Tandfonline. (2025). Full article: Aligning socio-technical systems: Rethinking AI adoption and digital transformation in SMEs. Information Systems Management. https://doi.org/10.1080/10580530.2025.2612175
21) Trilateral Research. (2025). Human-in-the-loop AI balances automation and accountability. https://trilateralresearch.com/responsible-ai/human-in-the-loop-ai-balances-automation-and-accountability
22) Vararean-Cochisa, D., & Crisan, E.-L. (2024). The digital transformation of the construction industry: A review. IIM Ranchi Journal of Management Studies, 4(1), 3–16. https://doi.org/10.1108/IRJMS-04-2024-0035

